

Optimal Disk Management Scheme for a Hypothetical OS in HPC Environments



CSE 4501 : Operating Systems

Assigned to :

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Optimizing Disk Management for **High-Performance Computing Systems**: A Strategic Design Approach

Abstract :

This assignment proposes a disk management scheme customized for a high-performance computing (HPC) OS. HPC demands extremely high throughput, low latency, and maximum reliability from disk subsystems. The scheme selects a suitable disk scheduling strategy, which focuses on the algorithms that minimize seek time and improve overall I/O efficiency. The scheme also outlines mechanisms for disk formatting, space management, and data integrity to support heavy parallel workloads. The proposed solutions are evaluated against common alternatives, with each design choice justified using theoretical principles from the Operating Systems course.

1. Disk Performance Requirements in HPC Environments

Requirement	Explanation
High Throughput	HPC tasks are data-intensive; need to move large volumes of data fast ^[1]
Low-Latency	Rapid access to disk to avoid I/O bottlenecks ^[1]
Parallel I/O Support	Multi-threaded applications demand concurrent data access
Reliability	Data integrity must be preserved during long or intensive computations
Predictable Performance	Important for time-critical simulations or batch processing

2. Disk Scheduling Strategy :

Chosen Algorithm: C-SCAN

2.1 Justification:

- Consistent response time: Provides more uniform wait times than SCAN. ^[2]
- Better suited for real-time/batch processing in HPC where predictability is key. ^[2]
- Avoids starvation (unlike SSTF), and avoids frequent direction changes (unlike SCAN). ^[2]
- Works efficiently with SSD arrays or RAID configurations common in HPC. ^[2]

2.2 Why not other algorithms?

Algorithm	Reason not chosen
FCFS	Too simplistic; doesn't optimize head movement
SSTF	May starve distant requests, causing latency spikes
SCAN	Slightly less predictable than C-SCAN
LOOK/C-LOOK	Similar but adds complexity without significant performance gain in HPC contexts

3.Disk Formatting & Space Management

- Disk Formatting Strategy:
 1. Low-level formatting based on sector alignment and block size optimization (e.g., 4KB or 8KB blocks).
 2. Include multiple partitions to isolate system files, temp/swap, and high-priority I/O.
- Space Management:
 1. Use extent-based allocation instead of block-by-block, which **reduces fragmentation** & is efficient for large **sequential I/O** in HPC.
 2. Implement bitmap-based free space tracking for fast allocation/deallocation.

4. Reliability Mechanisms :

- **RAID Support:** Use RAID 6 or RAID 10 to balance speed with redundancy. ^[3]
- **Journaling File System (e.g., XFS/ZFS-like):**
Ensures recovery in case of crash/power loss.
- **Checksums & ECC:** To detect and correct corruption. ^[4]
- **Snapshot/Shadow Copy:** For long-running simulations—ensures rollback in case of failure.

5. Design Summary :

Component	Design Decision	Why best for HPC
Disk Scheduling	C-SCAN	Predictable, avoids starvation
Allocation Strategy	Extent-based + bitmap tracking	Efficient for large, sequential files
Formatting	Sector-aligned + block-optimized layout	Matches I/O behavior of HPC apps
Reliability	RAID, journaling, checksums	Protects against failure, recovers

6. Theoretical Justifications

- C-SCAN minimizes **disk head movement variance**—which is critical in systems with real-time or scheduled tasks. ^[2]
- Extent-based allocation reduces internal fragmentation, which increases read-ahead capability.
- Bitmap tracking is **O(1)** for search and allocation compared to linked-list or FAT-style.
- RAID & journaling ensure data integrity and recoverability—non-negotiable in long HPC jobs. ^[3]

References :

[1]<https://www.ibm.com/think/topics/hpc>

[2]<https://www.geeksforgeeks.org/c-scan-disk-scheduling-algorithm/>

[3]<https://www.geeksforgeeks.org/raid-redundant-arrays-of-independent-disks/>

[4]https://www.tutorialspoint.com/data_communication_in_computer_network/error_correcting_codes.htm